

Cover letter

30 August 2026

Editors npj Systems Biology and Applications

Dear Editors,

Please consider our Article, "Disease-blind reconstruction distinguishes reproducible interferon remodeling from unstable B-cell state assignments in systemic lupus erythematosus", for publication in npj Systems Biology and Applications.

The manuscript addresses a systems-biology problem that is central to single-cell disease research: whether conclusions remain stable when cell identity, composition and transcription are treated as distinct inferential layers. Using public systemic lupus erythematosus datasets, we combined disease-blind B-lineage reconstruction, biological-unit-aware inference, end-to-end assignment sensitivity and independent source-label-defined replication. The analysis preserves two prespecified negative gates rather than tuning them away: the broad B-cell partition remains an analysis scaffold after an antibody-secreting-cell overlap criterion failed, and corrected source-label-independent external remapping did not qualify for disease-effect estimation.

The main contribution is therefore an evidence hierarchy. Hard state assignments have explicit reproducibility and transfer limits, whereas a prespecified IFN/ISG process-level signal persists through uncertainty propagation and independent donor-level replication. This focus aligns with the journal's scope in single-cell systems biology, systems immunology and computational analysis of complex disease systems. Regulatory analyses provide convergent but observational context and are not presented as causal proof.

All analysed datasets are publicly available. Code, decision records, figure source data and complete statistical outputs are available through GitHub and the version-specific archive at <https://doi.org/10.5281/zenodo.22151739>. The work is original, is not under consideration elsewhere, and the authors declare no competing interests.

Thank you for your consideration.

Sincerely,

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